



NEWSLETTER

JUNE 2024



BRAND NEW FSAE TEAM

WHAT HAVE THEY DONE
IN AN ENTIRE MONTH?

WATER COOLING SYSTEM

POSSIBLY THE MOST
IMPORTANT SYSTEM

THE TEAM GETS THEIR FIRST

SPONSOR

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WHAT IS AUS RACING

AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers

to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

A WORD FROM OUR TEAM PRINCIPAL



Greetings,

It is with great honor that I announce AUS Racing, the newly formed Formula Student team from the American University of Sharjah. We have recruited engineers and volunteers who are passionate to compete in the international competitions of Formula Student.

Winning in Formula Student is only the hidden summit on this steep hiking journey. Every member will gain hands-on experience in their interested fields, learn to work in a team and push their imaginations to build a racecar from scratch. We hope to give students of any discipline and standing the opportunity to take part in the best learning

experience.

For this season we aim to lay the foundations at our university for decades of successful participation and the ability to produce increasingly competitive cars. Throughout this journey we wish to inspire future teams not only in AUS but the entire MENA region.

We are really excited about this expedition, and we invite you to join us. Your support and enthusiasm will certainly push the team towards success.

Join us in our journey to design and build a Formula Student car through our monthly newsletters. Enjoy the read!

-Saad Ibrahim Naikwadi
AUS Racing Team Principal

"IT'S ABOUT PEOPLE THEN MONEY. IT'S ABOUT PEOPLE THEN MACHINE."

-CLAUDE ROUELLE

OUR WHEELS

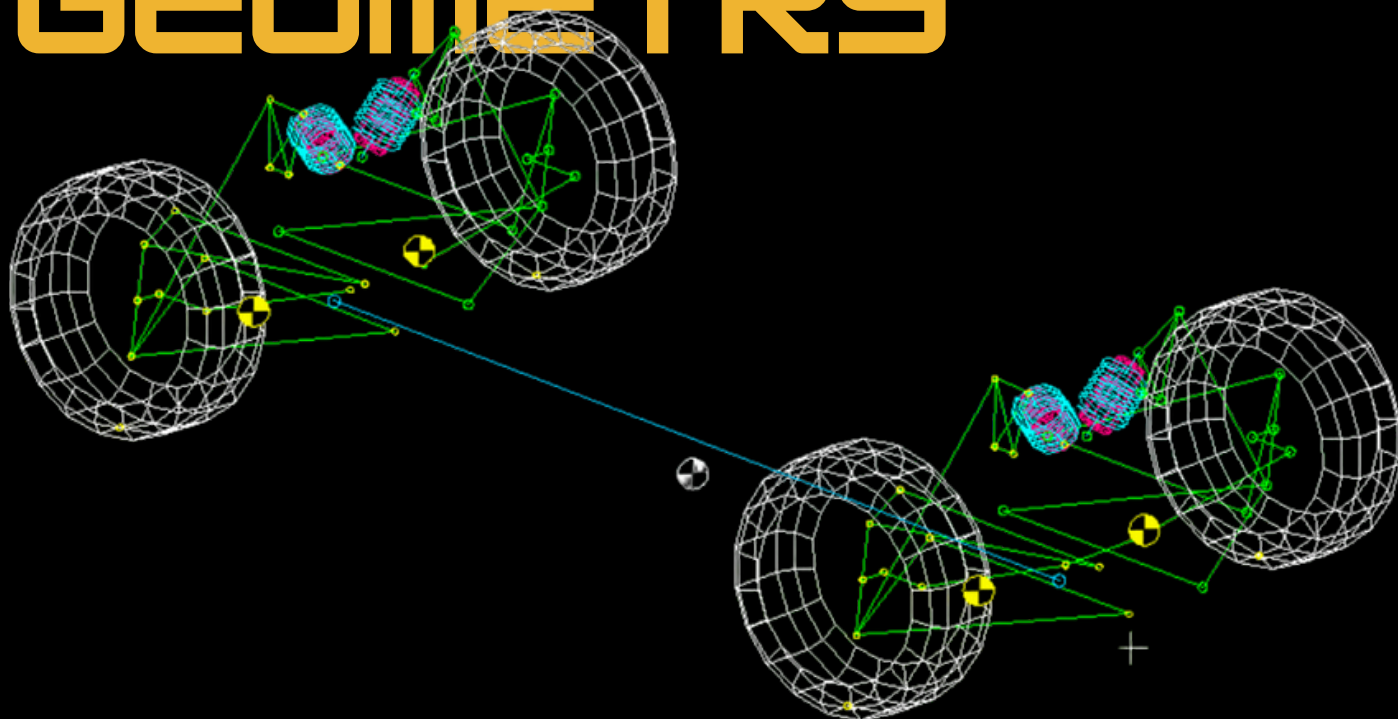


The tires are the most important part of the car. They are the first and only contact of the car with the track and are crucial in determining the performance of the car. While it was difficult to select tires without any prior data, the team was still able to find the best suited tires using data from extensive research and simulations.

The rims are an integral part of the wheel. Nothing sounds better than forged aluminum multi-piece rims. Lightweight customizable

rims are the ideal choice for a project like this. Selecting the tires and rims had to be the first step we took in the design process as they define everything from the positioning of the suspension to the dynamics of the car

SUSPENSION GEOMETRY



The suspension of the car has many interconnected factors. Improving each factor negatively influences at least two others. To approach such a problem we divided our design into six phases. Our first completed phase involves focusing on optimizing front view geometry, specifically, the camber gain when the car goes over a bump or rolls in a corner.

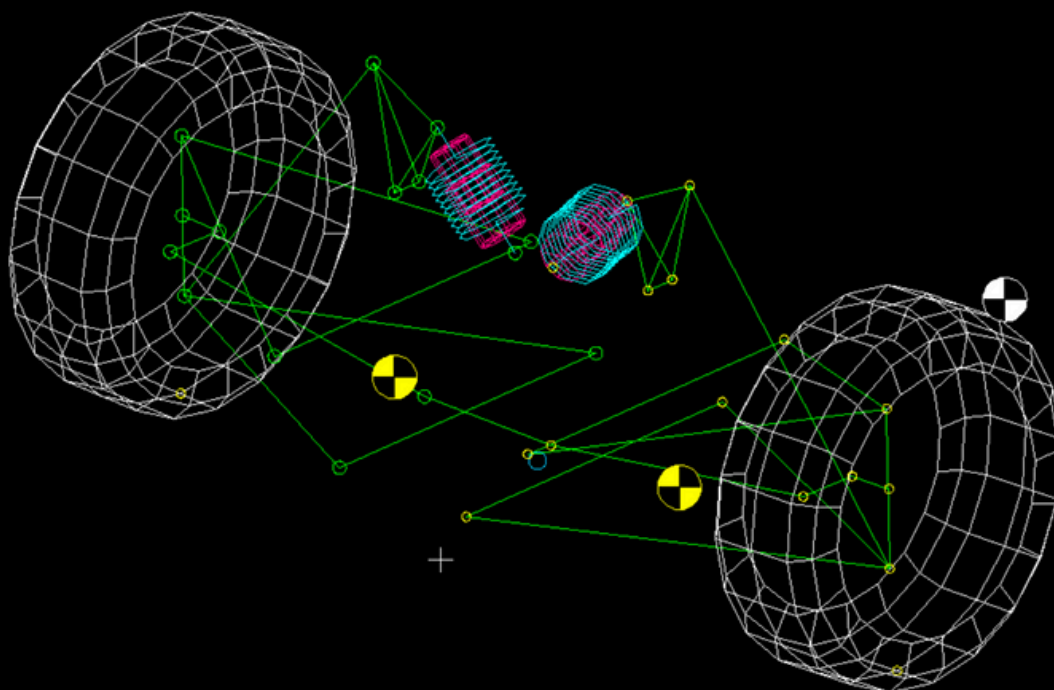
Negative camber is created by orienting the wheels at an angle such that the top of the wheels are angled inwards. Doing this helps

to increase the tire grip.

Camber gain is the characteristic of a wheel's camber angle to change as the wheel travels up and down. The lengths and positioning of the control arms in the front view determine how much the camber changes with wheel travel. Ideally, there should be no camber gain. However, this is not possible and reducing camber gain comes with numerous trade-offs which need to be balanced for optimal performance.

STEERING

GEOMETRY



Steering, although it sounds simple, is an extremely extensive study for the vehicle dynamics. Steering is heavily dependent on the suspension geometry of the car. However, similar to suspension, steering also has a variety of factors that need to be balanced and fine-tuned to squeeze the most performance out of the car.

It is impossible to tackle the

steering geometry design as a whole all at once. The design process has been divided into six iterative phases to make the design process easier and more efficient.

The first phase involved understanding and calculating the Ackermann geometry, a geometry that is crucial for increasing the grip of the tires and reducing the tire wear.

BATTERY TO WHEELS



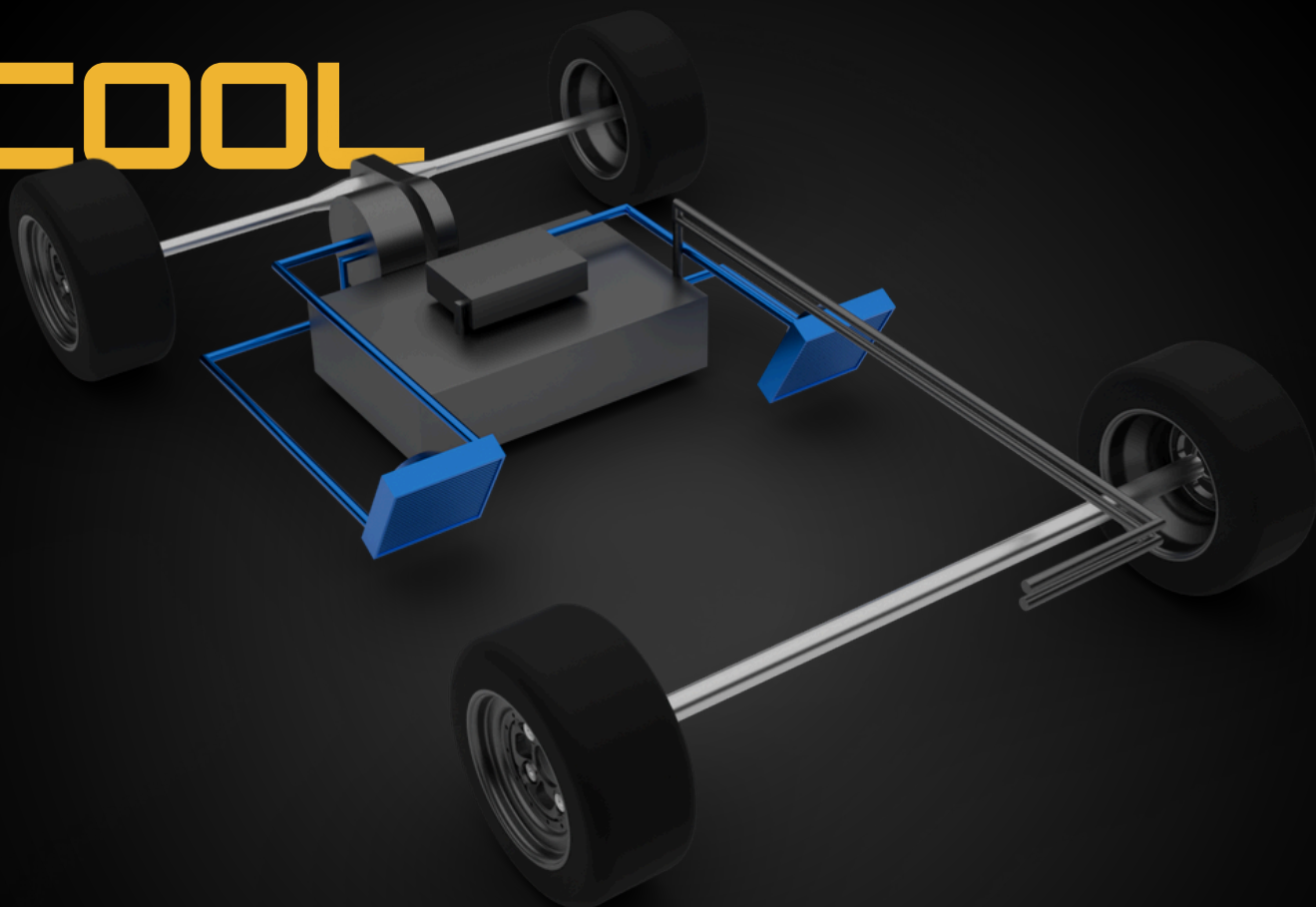
Getting power to the wheels is not as easy as it may sound. In fact, it is a very challenging task. As the inaugural year team we have decided to outsource our battery design and work with a supplier to design a regulations compliant battery and use the knowledge to design our own battery next year.

However, all the other intricate systems as required by the rules and for performance will keep the powertrain team busy.

Everything from master switches to regenerative braking will require a considerable amount of attention and hours in both hardware and software.

Running simulations has helped the team get a good idea of how the car's performance is affected by the battery capacity, motor, transmission ratio and a variety of other variables.

KEEPING IT COOL



Cooling is an extremely important system of the car. The smallest error in this system can lead to anything from something as mundane as the motor delivering less power than needed to the battery catching fire.

Our cooling system will incorporate two radiators on either side of the driver to dissipate heat from the water cooling system which will be used to cool the battery, motor, circuitry etc.

However, don't be fooled by what seems like a simple system on the surface. The circulation of water while reducing weight and keeping it effective will be a difficult balance. Not to mention the aerodynamic challenges introduced with the addition of two massive drag generating radiators placed directly behind rapidly rotating wheels which are trying to reduce the aerodynamic efficiency of the system!

MEET OUR NEW SPONSOR

The Automech Group has offered us a gracious sponsorship by helping us manufacture, fabricate and source raw materials. They are an extremely comprehensive machine shop and fabrication company in the UAE. Their strive for excellence in delivering to their clients and the esprit-de-corps shared by their employees is awe-inspiring for our team. They have cultivated innovation and productivity in the industry for over thirty years and we hope to walk in their footsteps.

Thank you Automech for your generous sponsorship!

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